

W4 — Sewer Network Design Pipeline

Methodology and Test-Boundary Results

Internal working document — design team

18 August 2026

Item	Value
Test area	551 ha
Network designed	2,137 chambers / 90.4 km
Peak flow at outfall	83 L/s (Qadf 3,070 m3/d)
Structural rules	loop-free PASS · one outlet per chamber PASS
Audit	2 residual violations
Code / tests	W4/py/sewnet — 44 pytest cases, Table-11 gate

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Executive summary

We built the sewer design pipeline, proved it end to end on the 551 ha test boundary, put it through a 21-agent adversarial review and a two-rule structural audit (no loops; one outlet per chamber), fixed everything both found, and the design now holds those rules with 2 marginal residuals out of 2,137 chambers. Here is the whole story in one page.

The pipeline takes four inputs — roads, classified plots, the 0.5 m terrain and a boundary — and produces a complete gravity network: 2,137 chambers, 90.4 km of pipe (95% DN200, stepping up to a DN600 outfall leg), designed inverts everywhere, about 11 seconds to run. Every one of the 2,987 loaded units (2,217 built plots, 522 planned, 248 unparceled buildings; 86 farms excluded per doctrine) lands on exactly one chamber — nothing silently dropped, mass balance closes exactly. Saturation flow at the outfall: Q_{adf} 3,070 m³/d, peak 83 L/s (Merrimack).

The hydraulics were never taken on faith. The Colebrook-White solver had to reproduce all nine minimum gradients of G203 Table 11 (plus/minus 5%) before it was allowed to size anything — it does, mean deviation under 2%. The adversarial panel then attacked the code and confirmed 11 real defects, all fixed: the spanning tree skipped 22 km of cross-street edges (now covered with a crest-manhole layout); capacity was computed on nominal diameter while OD-designated PVC-U bores smaller (now the true SDR34 bore); a broken bisection in the velocity-cap solver; drop structures measured against the wrong datum. 44 pytest cases lock all of it in.

The structural audit found what the graph checks had hidden (section 4): the tree gave every node one outlet, but 255 chambers sat within 3 m of another chamber — many at the same point — each with its own outlet. On the ground that is a two-outlet junction, exactly what the rule forbids. The pipeline now merges coincident chambers, re-derives the tree over the merged set (which also removes loops that merging exposes), and offsets every extra outgoing pipe so it starts at the next house connection or 10 m clear of the chamber — the SWNETWROK convention, which had not been adopted until it was called for.

The honest self-cleansing picture, on the table rather than under it: 95% of pipes cannot reach 0.75 m/s at saturation peak — physically inevitable on small residential branches — and comply through the guideline's own tractive-force alternative at $\tau = 1$ Pa, a pending assumption [GAP-9]. If NWS sets $\tau = 2$ Pa, 1,515 pipes need steeper slopes. That single number is the strongest argument for pinning τ down at the kickoff.

Three findings from the test area:

- **The outfall landed 733 m from where you expected** — the terrain's lowest boundary road node is at (450614, 2567397), south-center on the main corridor, z 351.2 m, not the west edge. If the real trunk connection is elsewhere, it is one config line and an 11-second re-run.
- **Your elevated-roads concern was justified:** 289 units (9.7%) could not gravity-connect at standard sewer depth. Deepening 211 chambers recovered all but 10, now flagged as local-solution candidates.
- **2 SLS pockets appeared** (28 nodes, about 5 properties total) — absorb-to-detail-design per rule 9. 382 drop structures (225 vortex-class) concentrate at wadi-bank crossings, exactly where detail design applies the G1-p85 rules.

Also delivered: T01 Rev 3 — the tutorial now teaches Colebrook-White (section 14), Table 11 derived step by step, every number verified.

What is proven: doctrine loads in, guideline-compliant network out, auditable and re-runnable, honest about its assumptions. What is not yet proven: SewerGEMS agreement (package and referee table await the ModelBuilder run), and behaviour at 36-zone scale with the finalised trunk. That is W5.

1. What the pipeline is

A re-runnable Python package (sewnet) that designs a gravity sewer network inside any boundary, given roads, loaded plots, terrain and an outfall or connection point. One config file holds the paths; criteria.py holds every design number with its PAM-GUD page reference — no number lives anywhere else in the code.

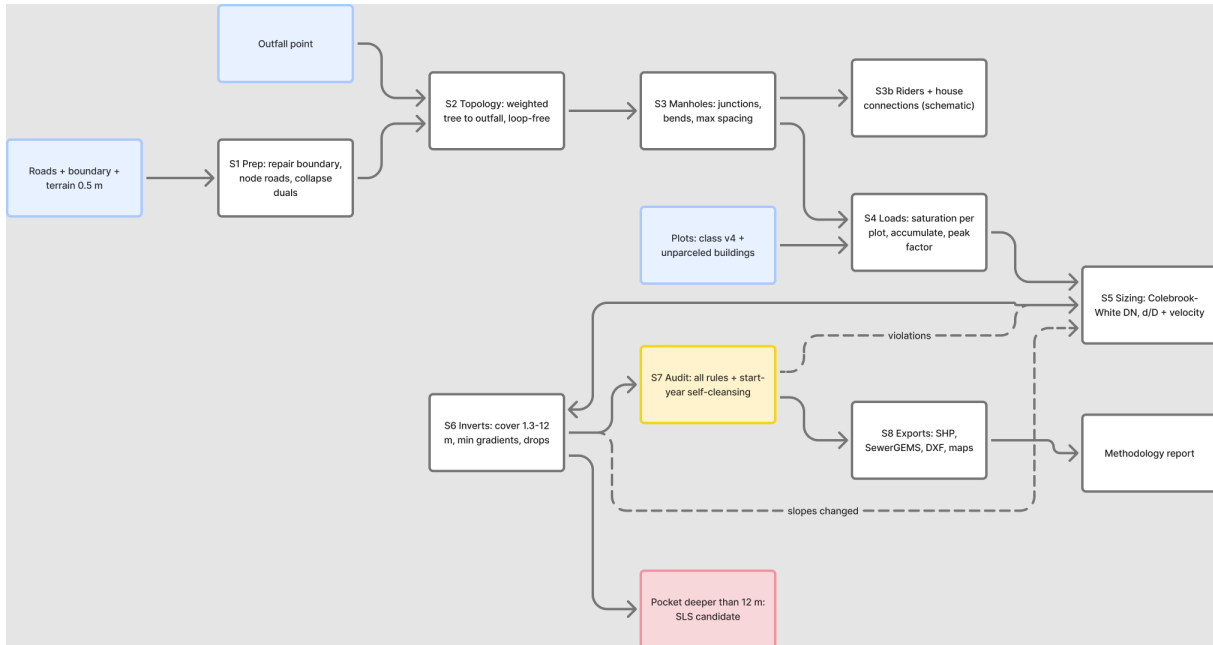


Figure 1 — Pipeline architecture. Blue = inputs, yellow = audit gate, red = SLS flag, dashed = iteration loops.

Stages, in one breath: repair and clip the inputs; node the roads and collapse dual carriageways; grow a loop-free tree toward the outfall (climb-penalised, arterial-preferring — the 'no alleys' lesson); add cross-street branches where loaded plots front an off-tree street; place chambers (junctions, bends over 45 degrees, 100 m spacing); merge coincident chambers and offset extra branch starts; load every plot at saturation; accumulate flows with peak factor and infiltration; size pipes and solve inverts together on the true bore; check every house can reach its chamber, deepening where roads are elevated; audit everything independently; export SHP, SewerGEMS, DXF and maps.

2. Hydraulic basis and how it is verified

Element	Basis	Verification
Capacity / velocity	Colebrook-White, $ks = 1.5 \text{ mm}$, $nu = 1.141\text{e-}6 \text{ m}^2/\text{s}$ (G203-p24/25/28), partial-full circular geometry, true internal bore (PVC-U OD-series derated to SDR34 ID; GRP nominal = ID)	Table-11 gate: reproduce all 9 minimum gradients at 0.75 m/s within 5% — passes at under 2% mean deviation; ID convention unit-tested
Minimum gradients	Steeper of Table 11 (p29) and tractive force $S_{min} = 2.33\text{e-}4 \times \tau^{1.23} \times Q^{-0.461}$ (p27, A9-corrected), plus a 40 mm total-fall guard per reach (p29 4.3.1)	$\tau = 1 \text{ Pa}$ tagged [GAP-9]; Q floored at Mara's 1.5 L/s minimum design flow — at the floor tractive is about Table 11 DN200: the methods meet
d/D limits	at or below 0.65 ($D \leq 350$) and 0.50 ($D > 350$) at peak (p27 Tab 10)	enforced in sizing, re-checked in audit on the true bore
Velocity band	at least 0.75 m/s at peak or tractive-compliant (p26-27); at most 3.0 m/s via a slope cap whose surplus fall becomes a designed drop	audit plus the transparency statistics in section 6

Element	Basis	Verification
Loads	Doctrine 2: every plot at saturation, $6.0 \times 171.3 \text{ l/c/d} = \text{about } 1.03 \text{ m}^3/\text{d}/\text{plot}$; farms zero; infiltration 720 L/d/km unpeaked; PF Merrimack (mandatory above 100 properties), held at its 100-property value below; Peltier comparison held the same way	mass balance to the outfall closes exactly; unknown CLASS values can never vanish silently

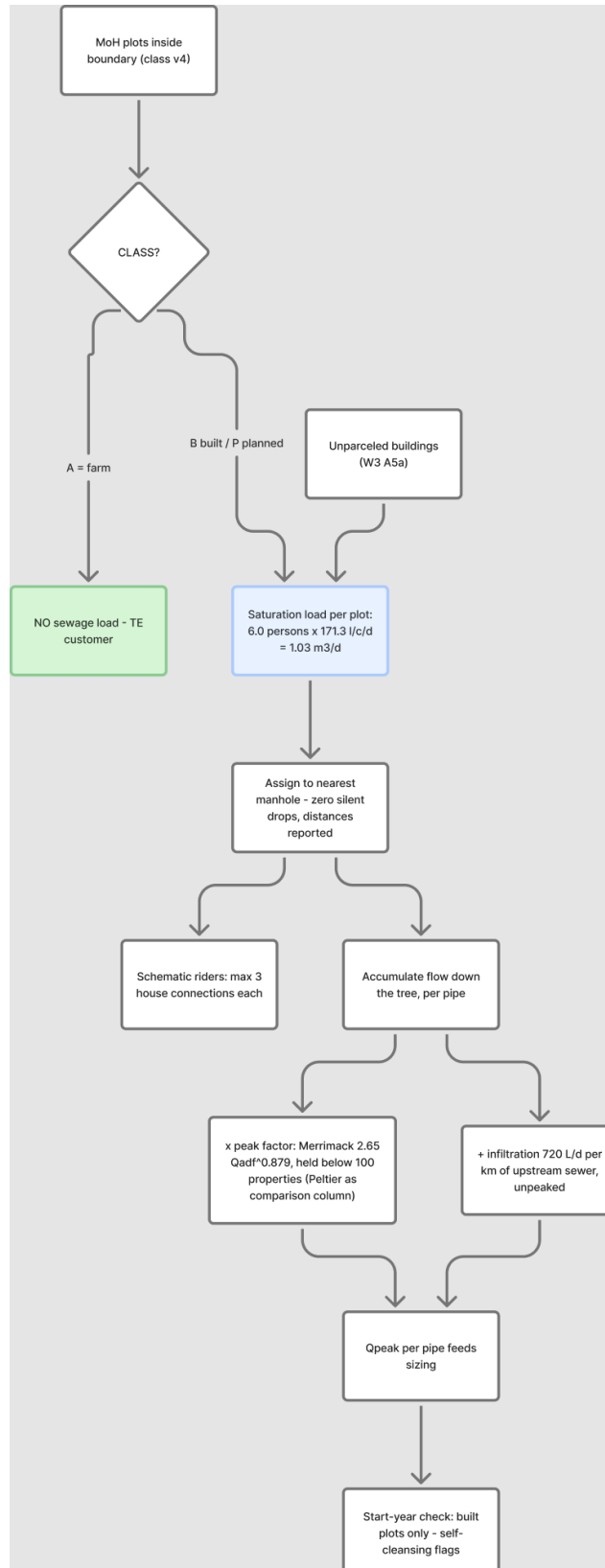


Figure 2 — Load allocation chain.

3. How the solver designs a pipe

Two passes per reach, iterated until no diameter changes (2 iterations sufficed; oscillation between adjacent DNs is detected and broken upward; a final lay pass always leaves inverts consistent with final diameters).

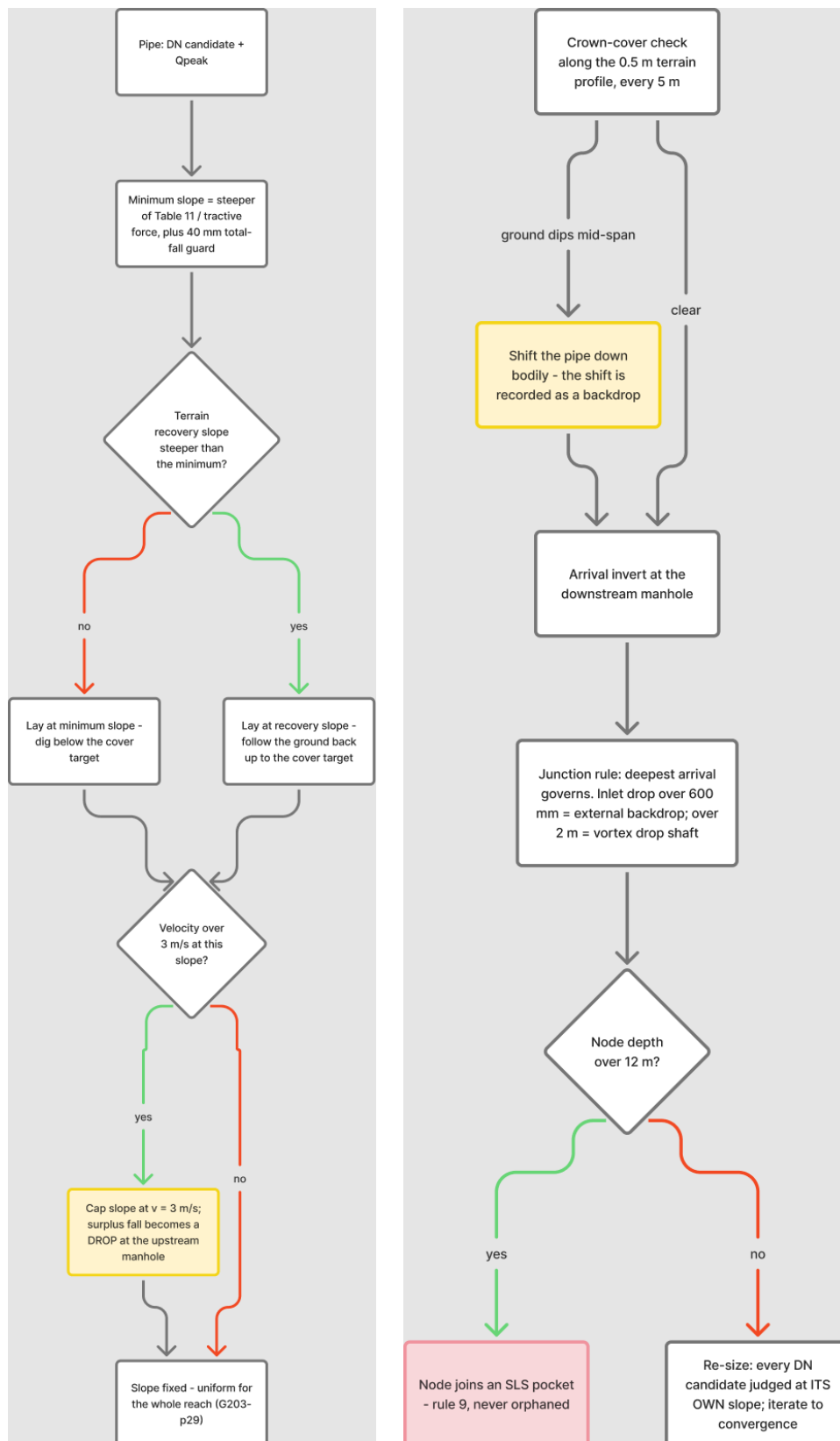


Figure 3 — Solver logic: step 1 sets the reach slope, step 2 resolves profile, junction and depth.

The details that matter:

- **Uniform slope per reach (p29)** — one straight line between chambers, never kinked.
- **The chamber datum is the outgoing invert** — chamber depth is construction depth, and every inlet drop is measured inlet-invert minus outgoing-invert, so velocity-cap surpluses and cover shifts combine into the drop they physically are. Over 600 mm = external backdrop, over 2 m = vortex shaft (p30); the audit re-derives every drop independently, and a missing or misclassified record is a violation.
- **Mid-span cover on real terrain:** every reach profiled at 5 m on the 0.5 m VRT; a reach riding above dipping ground is shifted down bodily, the shift surfacing as a recorded drop.
- **Sizing without the ratchet:** every candidate DN is judged at its own governing slope (no-oversizing, p29), on its true bore, with velocity-infeasible candidates skipped explicitly.
- **Over 12 m depth becomes an SLS pocket (p33 + rule 9)** — never a silent orphan.

4. Loop-free, and one outlet per chamber

Both rules are enforced constructively, not checked after the fact, and both are verified independently on the final output.

No loops. The collection network is a spanning tree by construction: a cost-weighted shortest-path tree to the outfall, where each chamber's single outgoing pipe is its next hop. Loop-closing street edges can never become pipes in that step. Final output: 2,137 chambers, 2,136 pipes, one connected component, acyclic — the pipe count being chambers-minus-one is the arithmetic signature of a tree.

One outlet per chamber. A junction takes any number of inlets and exactly one outgoing pipe. The catch the graph checks missed: two separate chambers can sit at the same physical point (road noding rounds to the centimetre, and the cross-street augmentation planted branch heads next to existing junctions). Each had one outlet in the graph; together they were a two-outlet junction on the ground. The fix, in order:

- **merge every cluster of chambers within 3 m** — they are one structure, which makes the hidden fan-outs visible as chambers with two outgoing pipes;
- **re-derive the tree over the merged pipe set**, because merging can also close a loop (two chambers at one point may additionally be linked by a path). This restores one-outlet-and-no-loops in a single step and marks the leftover pipes as loop-closers;
- **offset every leftover pipe so it starts clear of the chamber:** at the next house connection along its own alignment, or 10 m when that connection sits nearer — the SWNETWROK FANOUT_GAP_M convention. A branch whose pipe is too short to keep a clear start is dropped and reported (its street is served from the far end);
- **slide any remaining branch head clear of neighbouring chambers**, iterating until nothing moves.

Test-area result: 255 coincident chambers merged, 125 fan-outs resolved, 357 branch starts offset, 90 branches dropped as served-from-far-end. Independent verification on the exported network: loop-free PASS, one-outlet PASS, 568 of 569 branch heads at least 10 m clear, one chamber pair at 2.78 m instead of 3.0 m. Those residuals are reported rather than hidden — both are local layout details for detail design.

The same checks now run in the audit every time (mh-clearance, head-offset, one-outlet), with a unit test on a synthetic two-chambers-at-one-point case, so this cannot regress silently.

5. The house-connectability check

Roads are locally elevated for flood protection and underpasses, so houses can sit below the sewer. For every loaded unit: plot ground (0.5 m terrain) minus 0.6 m outlet depth must reach its chamber's invert with 2% fall over the connection distance. Failures raise a deepening requirement on that chamber (capped 0.5 m short of the 12 m limit) and the invert solve re-runs; anything still failing is flagged for a local solution. Test area: 289 flagged, 211 chambers deepened, 10 residual.

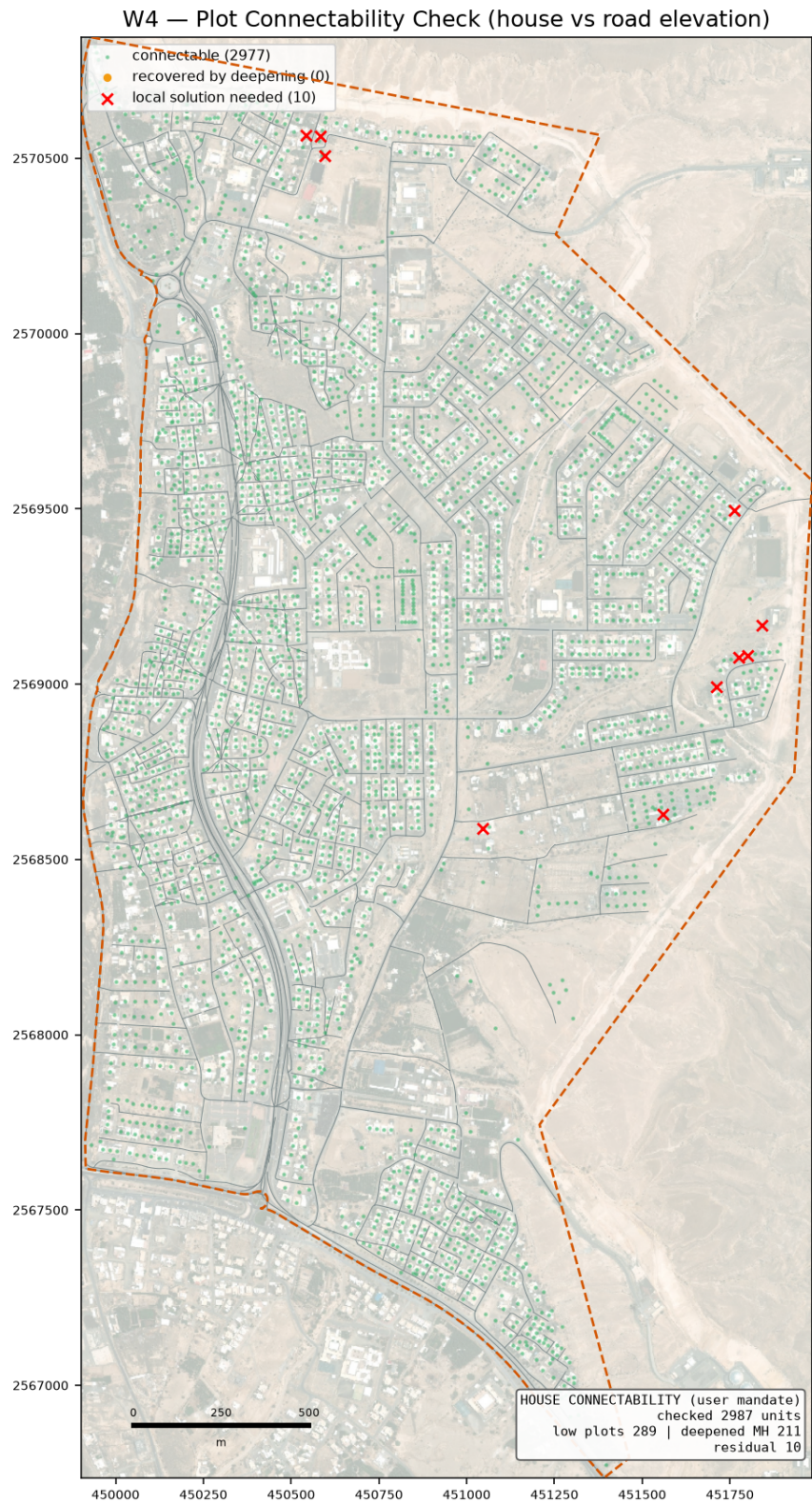


Figure 4 — Plot connectability: green connectable, amber recovered by deepening, red needs a local solution.

6. Test-boundary results

Quantity	Value
Area / roads / network	551 ha / 98.7 km roads / 90.4 km sewers (22.22 km from cross-street augmentation; 0.56 km of unloaded streets deliberately unsewered)
Chambers / pipes	2,137 / 2,136 (single tree, one outfall — chambers minus one)
Diameters	DN200 86.163 km · DN250 1.689 km · DN315 1.753 km · DN400 0.286 km · DN500 0.386 km · DN600 0.082 km
Loaded units	2,987 (built 2,217 + planned 522 + unparceled 248); farms excluded 86
Outfall	(450614, 2567397), z 351.24 — 733 m from user expectation
Qadf / Qpeak at outfall	3,070 m3/d / 83 L/s (PF merrimack; Peltier column in the shapefiles)
Structure rules (section 4)	255 coincident chambers merged · 125 fan-outs resolved · 357 branch starts offset · 90 branches dropped
Depths	max 15.1 m (inside an SLS pocket); gravity network otherwise at or under 12 m
Drops	382 designed structures; 225 vortex-class (over 2 m), concentrated at wadi-bank crossings
SLS pockets	2 (about 5 properties total, absorb per rule 9)
Low plots	289 flagged, 10 residual after deepening 211 chambers
Audit	2 violations — structural residuals only (one chamber pair at 2.78 m vs the 3.0 m clearance; one branch head at 9.1 m vs the 10 m offset); everything else clean, including independent drop re-derivation
Self-cleansing transparency	2,038/2,136 pipes below 0.75 m/s at saturation peak — compliant via tractive at tau=1 Pa [GAP-9]; 1,515 would need redesign at tau=2 Pa; start-year flags 2,082, all tractive-compliant
Runtime	about 11 s design + 4 s exports

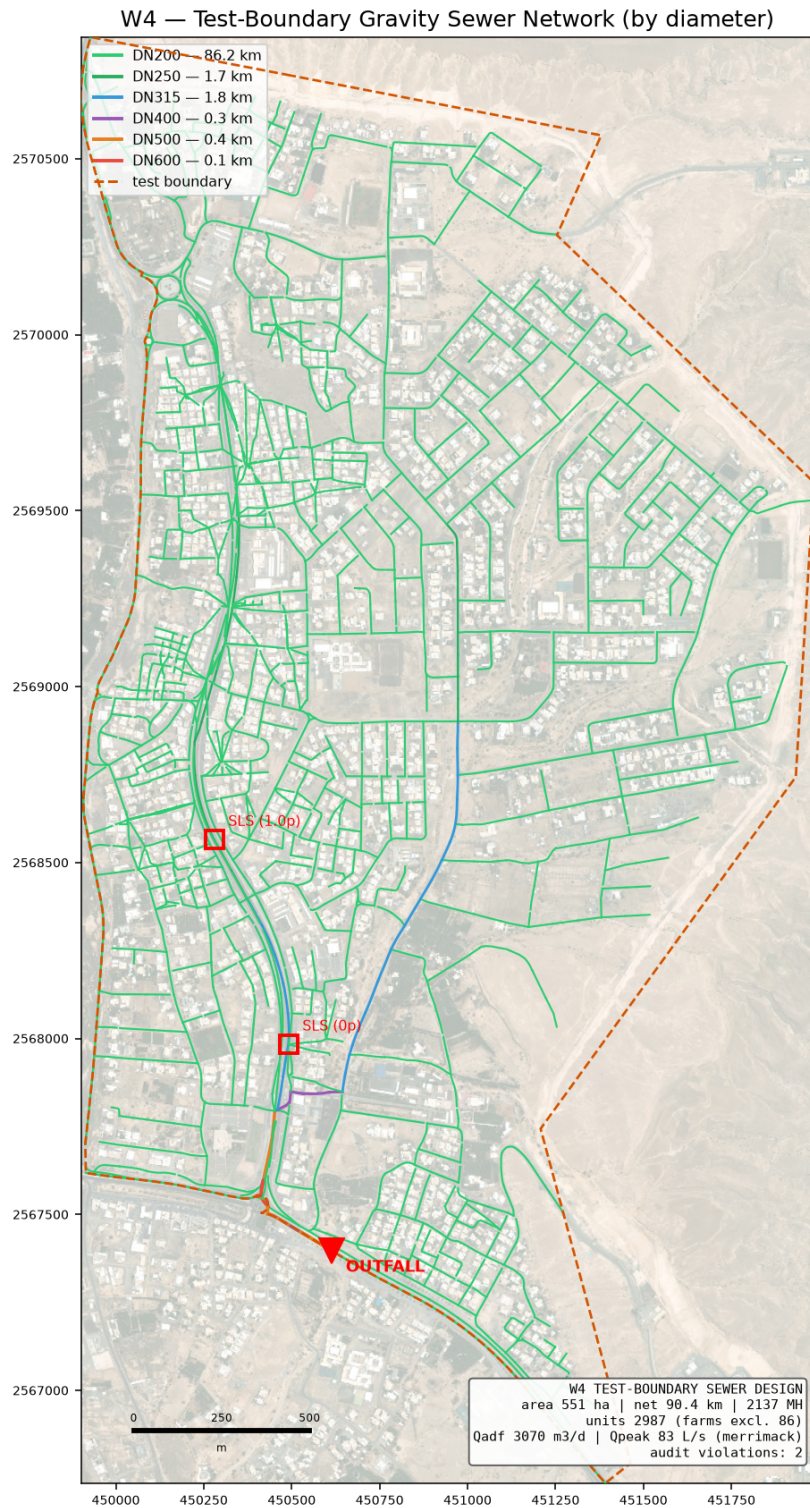


Figure 5 — Designed network by diameter, with outfall and SLS candidates.

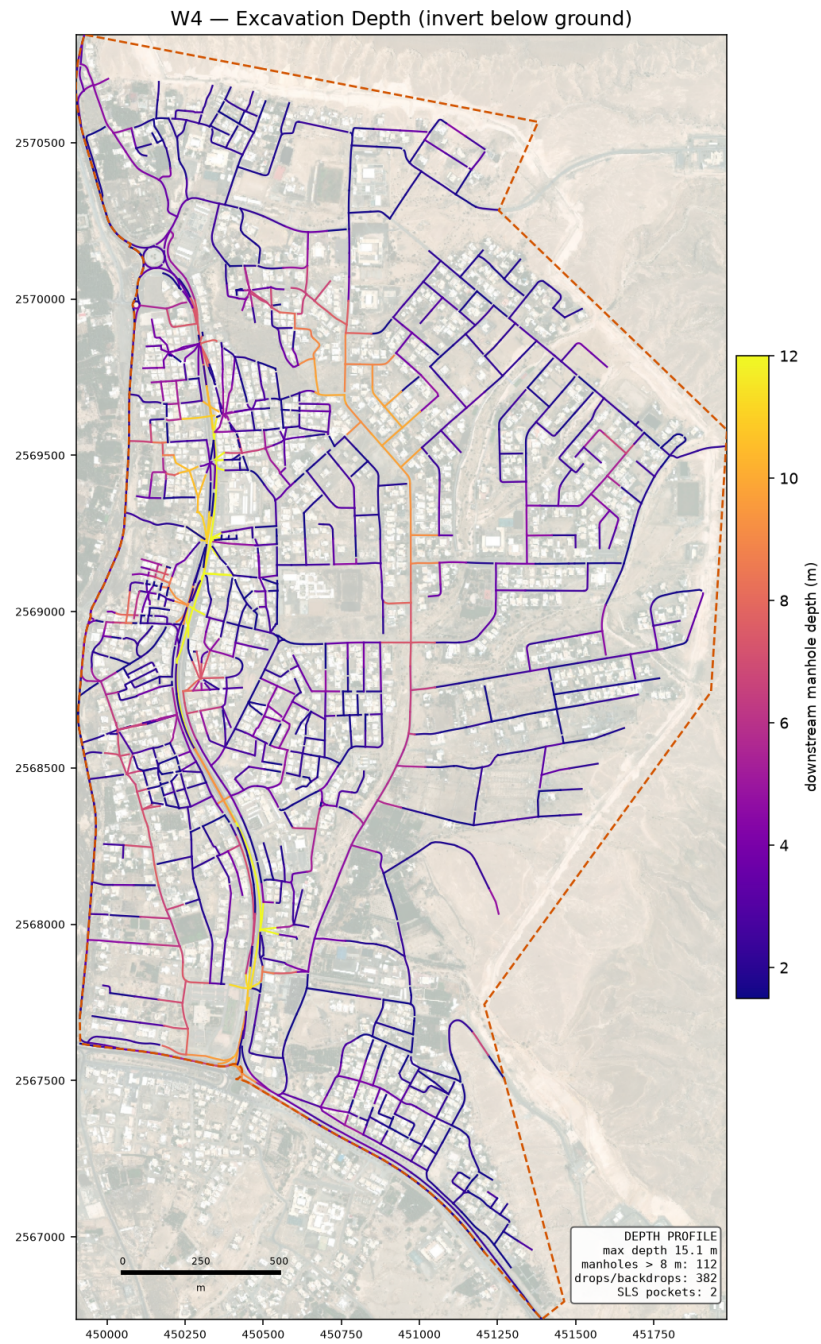


Figure 6 — Excavation depth (invert below ground).

7. SewerGEMS package and referee protocol

W4/sewergems/ holds MANHOLES, CONDUITS and OUTFALL shapefiles built to the Bentley-documented ModelBuilder mappings (explicit START_ND/STOP_ND plus vertex-snapped geometry digitised upstream to downstream, elevations not depths, numeric mm diameters), LOADS.xlsx (pattern-based rows, L/s per chamber), and IMPORT_PROCEDURE.md with the known traps — the 'Set Invert to Start/Stop Node = False' global edit being the one that silently deletes drop manholes. Manning n is exported as 0.013, the ks = 1.5 mm equivalent: the Tab-8 range 0.009-0.011 conflicts with the ks mandate and would show roughly 30% phantom capacity, so that conflict is flagged as an NWS kickoff item.

After the model run, paste SewerGEMS discharge, velocity and d/D into REFEREE_pipes.csv; any pipe deviating more than 5% from our columns is an open investigation. The design is not verified until the two engines agree — deliberately a separate run, not a self-check.

8. Assumptions register

Tagged in criteria.ASSUMPTIONS with the same wording, and reported in every deliverable:

Assumption	Basis / exposure
tau = 1 Pa	GUD-203 gives no numeric design tractive stress [GAP-9]; largest redesign exposure (1,515 pipes at tau=2)
Tractive Q-floor 1.5 L/s	Mara simplified-sewerage minimum design flow; unfloored the formula demands unbounded slopes as Q approaches 0
PVC-U wall class SDR34/SN8	true bore for hydraulics; actual class per PAM-SPC-207 pending
OR 6.0, 1 property per plot	GAP-5 (NCSI housing units missing)
Plot outlet 0.6 m, 2% connection fall	method choice for the connectability check
40 m frontage rule	an off-tree street gets a sewer when a loaded unit lies within 40 m
10 m branch-start offset, 3 m chamber clearance	layout conventions from SWNETWROK and the user's rule, not PAM-GUD values
Chamber size ladder	02 has no size table; GUD-203 4.4 re-extract pending (no hydraulic effect at concept)
Rider discharge to nearest chamber	02 silent on the discharge point
Gravity in-road position	taken from the force-main clause (p51, A9) as an inference
PF held below 100 properties	G1-p71 prescribes no formula there; Peltier held the same way
Infiltration unpeaked	add-order not stated in GUD-201 — kickoff item

9. Limitations and what changes at full scale

- **The trunk is user-finalised (settled 2026-08-18)** — the pipeline designs subnetworks into the given connection points, which are config entries, not structure.
- **One outfall per run today;** multi-connection territory competition is the W5 structural addition.
- **Riders are schematic;** junction losses ignored (normal-depth hydraulics) — both noted for the SewerGEMS comparison.
- **The 0.5 m terrain is concept-grade for inverts near the 0.75-1.0 mm/m minimums (G1-p36);** flat trunk profiles at scale need survey-grade data — a registered data request.
- **tau = 1 Pa carries the largest redesign exposure** — pin it at kickoff.

10. Adversarial review and structural audit

A 21-agent skeptic panel attacked the hydraulic core after the first audit-clean run: four attack lenses (Colebrook-White implementation, solver clause compliance, load and audit doctrine, executed edge cases) followed by independent verification of every raw finding. 17 raw findings, 11 confirmed, 11 fixed, 6 refuted. The headline four were cross-street coverage, the true PVC bore correction, the drop-datum correction, and the velocity-cap bisection repair.

A second, user-driven audit then checked the two layout rules directly against the run output: no loops (held) and one outlet per junction (held in the graph, broken on the ground — see section 4). The 10 m branch-start offset had not been implemented at all; it now is, with permanent audit checks and a unit test. Both audits are recorded in W4/docs/REVIEW_FINDINGS.md.